

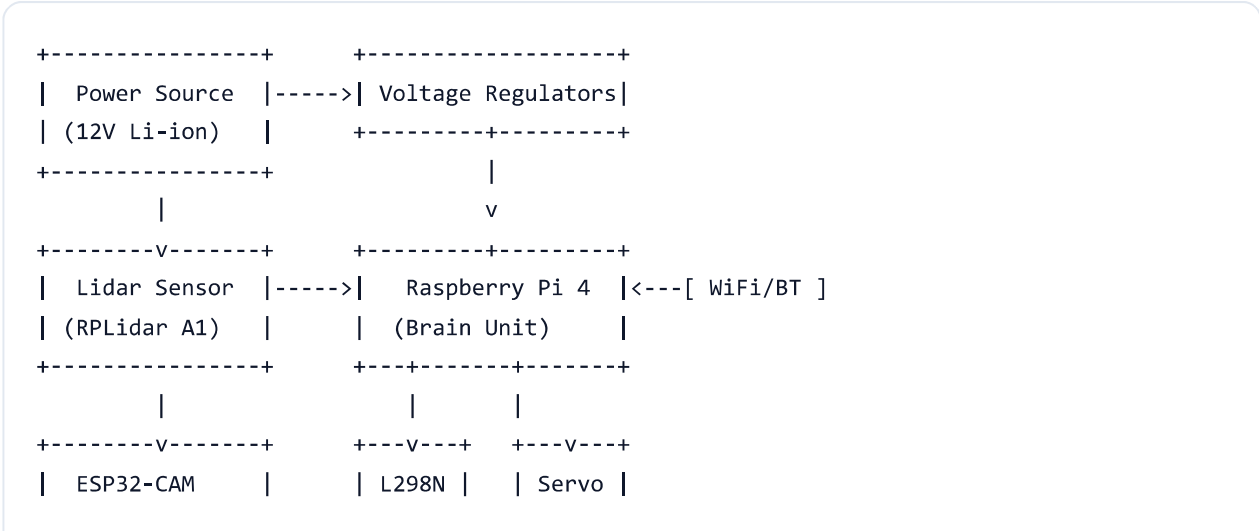
AI ROBOTICS PROJECT

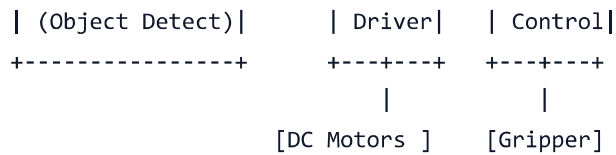
Autonomous AI-Driven Smart Waste Collection Robot

Abstract

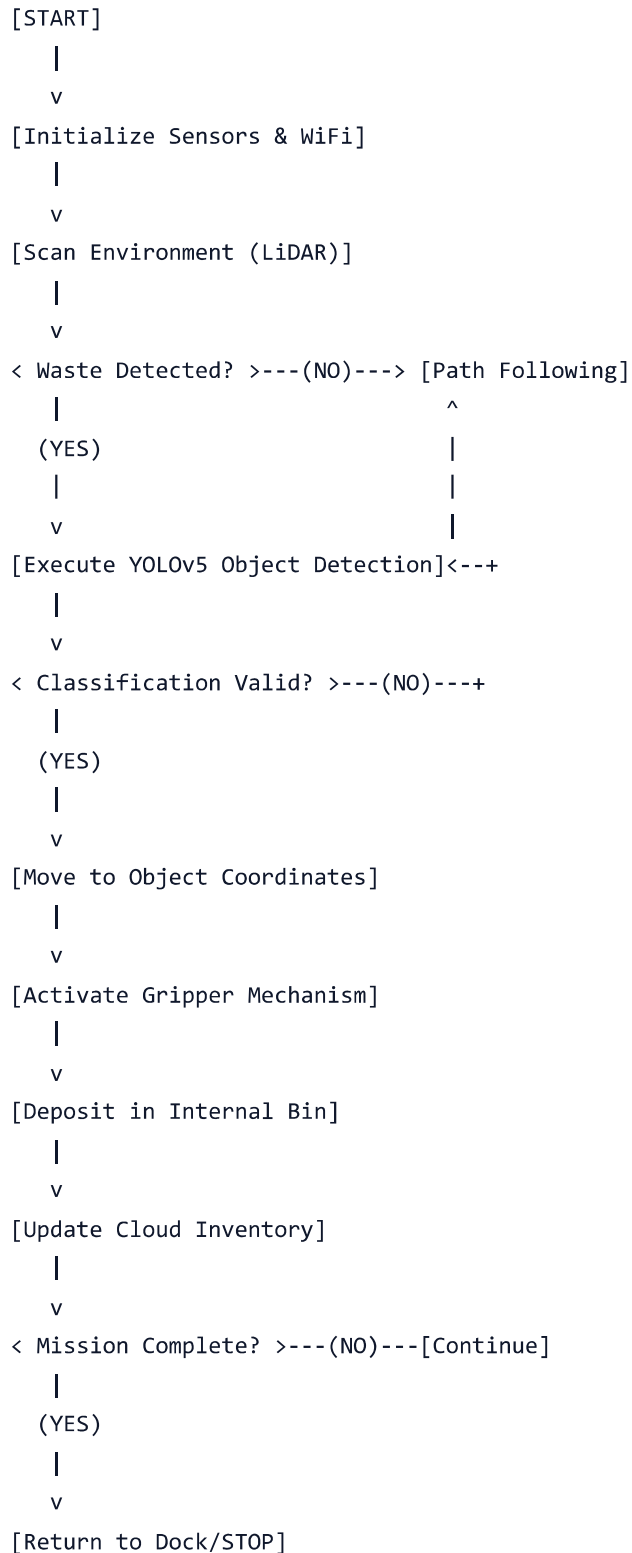
This project presents the design and implementation of an Autonomous AI-Driven Smart Waste Collection Robot, engineered to optimize sanitation in confined environments. The system utilizes a Raspberry Pi 4 as the primary controller, integrating a LiDAR-based navigation system for obstacle avoidance and mapping. Waste detection is achieved through a high-definition camera module utilizing a customized TensorFlow Lite model for real-time classification of recyclable versus non-recyclable materials. The mechanical structure features a differential drive chassis powered by high-torque DC motors and an automated gripper system for collection. For power management, a specialized Li-ion battery pack with a protection circuit is employed. Integration of an ESP32-CAM module facilitates secondary vision tasks and wireless data transmission to a centralized monitoring dashboard. The result is a robust, mobile robotic platform capable of autonomous path planning, intelligent object recognition, and efficient waste retrieval, significantly reducing the human labor requirement in waste management processes.

Block Diagram





Flow Chart



Hardware Required

Raspberry Pi 4 Model B (4GB)	Primary microprocessor for high-level computation, AI inference, and system coordination.
RPLidar A1M8	360-degree Laser Range Scanner for SLAM-based navigation and obstacle mapping.
ESP32-CAM	Secondary microcontroller with integrated camera for localized visual recognition and streaming.
L298N Motor Driver	Dual-channel H-bridge module for controlling the speed and direction of twin DC motors.
MG996R Servo Motors	High-torque metal gear servos for the operation of the mechanical gripper arm.
12V 300RPM DC Gear Motors	High-torque drive motors providing locomotion for the 4-wheel robot chassis.
HC-SR04 Ultrasonic Sensor	Short-range distance sensors for redundant proximity detection and safety braking.
12V 3000mAh Li-ion Battery	High-capacity power source with integrated Battery Management System (BMS).

Software & Tools

Raspberry Pi OS (64-bit)	Linux-based operating system for managing hardware resources and libraries.
Python 3.9	Primary programming language for logic, sensor integration, and motor control.
TensorFlow Lite	Optimized deep learning framework for real-time waste classification on the edge.
OpenCV	Computer vision library for image preprocessing, filtering, and contour detection.
ROS (Robot Operating System)	Middleware for handling communication between sensors, actuators, and navigation stacks.
RPLidar SDK	Driver and programming interface for processing LiDAR point cloud data.

Flask	Micro-web framework used to host the remote monitoring and teleoperation dashboard.
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Additional Components / Modules

4WD Robot Chassis	Durable acrylic or aluminum base plate designed to house all mechanical and electronic parts.
Mechanical Gripper Kit	2-DOF claw mechanism designed for grasping small refuse items like bottles and cans.
LM2596 Buck Converter	Step-down voltage regulators to provide stable 5V power to the Raspberry Pi and ESP32.
OLED I2C Display	Miniature 0.96 inch screen used for displaying system status, IP address, and battery levels.
Logic Level Shifter	Bi-directional module to safely interface 5V sensors with the 3.3V GPIO pins of the Raspberry Pi.

Key Learning Outcomes

- Design and integration of autonomous navigation systems using LiDAR and SLAM algorithms.
- Implementation of real-time object detection models using TensorFlow Lite on edge devices.
- Development of hierarchical control systems bridging high-level Linux OS with low-level motor drivers.
- Optimization of power management and wireless communication for mobile robotic platforms.
- Application of mechanical design principles for effective object manipulation and waste retrieval.